Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

Spring 2026

Problem Set #1

Posted: Monday, Jan 26 **Due: Tuesday, Feb 3, 11:59 pm**

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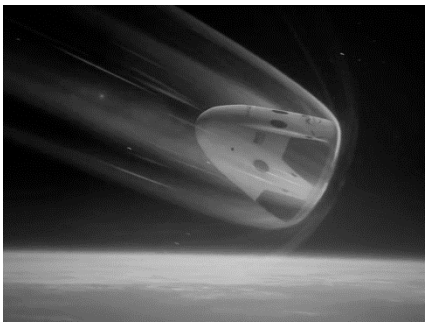
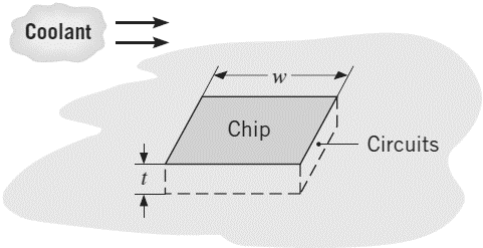
1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.



Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation $G_s = 750$ W/m^2, ambient temperature $T_\infty = 300$ K, surrounding temperature $T_{surr} = 300$ K, and heat transfer coefficient of air flow $h = 10$ W/(m$^2\cdot$ K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

Problem #1

Given: Spacecraft outside atmosphere
Spacecraft travelling through atmosphere

Find: Dominant heat transfer modes

a) Travelling through space, the spacecraft's dominant form of heat transfer is radiation, as there is no wind but the spacecraft is exposed to strong radiation waves from the Sun.

b) As the spacecraft enters the atmosphere, the dominant form of heat transfer is convection (although it experiences both convection and radiation). This is because there is moving fluid as it moves through the atmosphere.

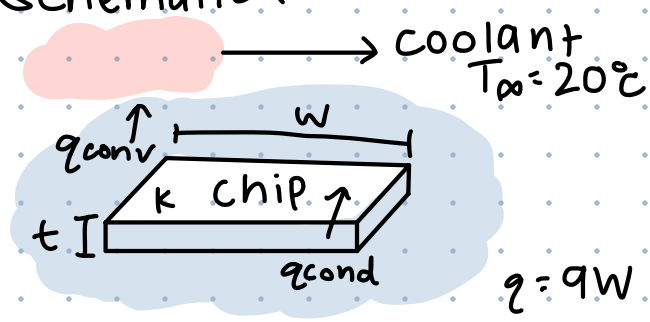
#2

Given: $k = 150\ W/m\cdot K$
$w = 5\ mm$
$t = 1\ mm$
$P = 9\ W$
$T_\infty = 20°C$

Side & Back thermally insulated
Front surface exposed to convective flows

Find: a) Steady State temp. difference between back and front
b) If the front $T_f \geq 80°C$, find the minimum heat transfer coefficient
Compare against Table 1.1

Schematic:



Assume: Steady State
Side & Back insulated
Negligible radiation
Negligible ΔPE, ΔKE

Analysis:

a)

$$q''_{cond} = \frac{k\Delta T}{l} \rightarrow q_{cond} = \frac{kA\Delta T}{l} = \frac{kW^2\Delta T}{t} \rightarrow \Delta T = \frac{qt}{kW^2}$$

$$\Delta T = \frac{(9W)(1\times10^{-3}m)}{(5\times10^{-3}m)^2(150W/m\cdot K)}$$

$$\Delta T = 2.4\ K$$

b) $q_{conv} = hA\Delta T = hA(T_{surf} - T_{surr})$ $\quad h = \frac{q}{W^2(T_{surf} - T_\infty)}$

$$h = \frac{9W}{(5\times10^{-3}m)^2(80-20°C)}$$

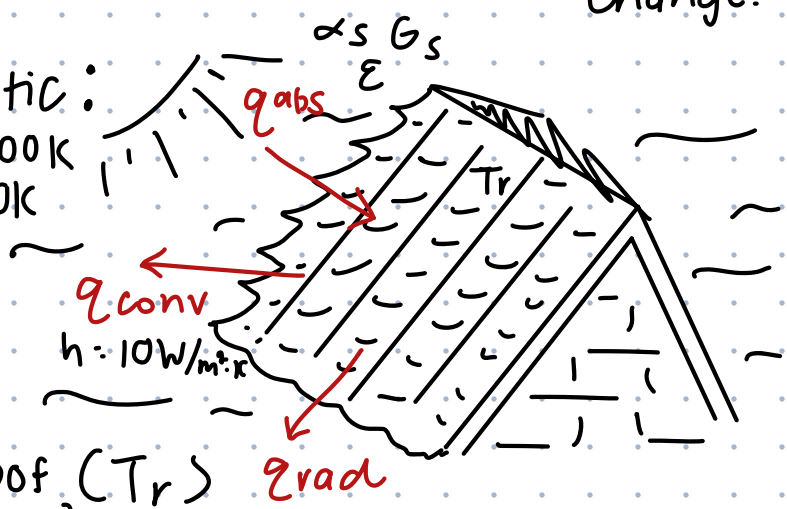
$$h_{min} = 6000\ W/m^2\cdot K$$

b)
(cont.) This coefficient is larger than expected for free convection and gaseous forced convection. It could work if the cooling fluid is a liquid (forced convection for liquids has a range of 100 - 20,000). Other methods could involve convection with phase change.

Problem #3

Given: $\varepsilon = 0.18$
$\alpha_s = 0.76$
$G_s = 750\ W/m^2$
$T_\infty = 300\ K$
$T_{surr} = 300\ K$
$h = 10\ W/m^2 \cdot K$

Schematic:



Find: a) Temperature of the roof (T_r) Would a cat be happy?
b) Discuss the dominant heat transfer mode

Assume: Bottom is thermally insulated
Roof can be treated as a grey body
Steady State

Analysis:

a) $q'' = \varepsilon\sigma(T_s^4 - T_{surr}^4)$ $\quad G_{abs} = \alpha G$ $\quad q''_{conv} = h(T_s - T_{surr})$

Steady State $\rightarrow \Delta q'' = 0$, $q''_{abs} = q''_{conv} + q''_{rad}$

$$[\alpha G = h(T_r - T_{surr}) + \varepsilon\sigma(T_r^4 - T_\infty^4)]$$

can graph to solve

$$(0.76)(750) = 10(T_r - 300) + (0.18)(5.67\times10^{-8})(T_r^4 - 300^4)$$

$$T_r = 350K$$

The cat would be unhappy as this would be ~170°F and burning hot.

Plugging in:

b) $h(T_r - T_{surr}) = (10)(50) = 500\ W/m^2$
$\varepsilon\sigma(T_r^4 - T_\infty^4) = (0.18)(5.67\times10^{-8})(350^4 - 300^4) = 70.5\ W/m^2$

The dominant heat transfer mode is convection, and based on h, likely free gaseous convection. This makes sense.